CLI CALCULATOR PROJECT

Project Debriefing Report

1. Introduction

This report summarises the outcome of Project 1, the Command Line Interface (CLI) Calculator. It records what the team achieved, the challenges it encountered and how they were addressed, the areas where the team can improve, what each member gained, and the plan for the next project.

The report is organised as follows:

Section	What it covers
2. Achievements	New skills learned and the successes of the project.
3. Challenges Faced	The difficulties the team ran into, how they were resolved, and how they will be handled in future projects.
4. Areas for Improvement	Recommended changes for future projects, for each member of the team.
5. What We Gained	What each team member gained from the project.
6. Next Steps	The next project and how members can prepare for it.

2. Achievements

- Delivered a fully functional, runnable program.
- Several members completed their first project.
- Worked effectively as a team.
- Learned what functions are and why they are important.
- Learned to use GitHub Desktop.

3. Challenges Faced

The table below lists each challenge together with the action taken, or planned, to address it.

Challenge	Resolution and planned action
Uneven Python experience	More experienced members taught those with less experience. For future projects, the team will train using coding practice sites such as Codewars and W3Schools.
GitHub conflicts	Future projects will use pull requests and separate branches, rather than working directly on the main branch.
Team coordination	A job file was used to keep everyone aligned, but it proved inefficient: it was easy to forget to update, and every update required a commit. The team will move to shared online documents to improve communication.

4. Areas for Improvement

The following changes are recommended for future projects.

- **Activity.** Every member should be more active in the group chats and begin work proactively, rather than waiting for a direct message. The team should also prepare to use Discord.
- **Reading project documentation.** Members should read all material provided, such as the design document, without waiting to be asked by the team leader. These documents contain the details and instructions the team must follow, and members who do not read them fall out of step with the rest of the team.
- **Delegation.** The team leader (Thisew) could have delegated more work to members who had finished their tasks. As a result, some members missed the opportunity to gain valuable programming experience.
- **Welcoming new members.** The team found it difficult to bring in members who joined mid-project, assign them work and bring them up to speed. Next time, we recommend giving one existing member responsibility for introducing the project to newcomers.

5. What We Gained

Each member records below what they personally gained from working on this project.

Member name	What I gained
Thisew	I gained hands-on experience leading a team and collaborating with others on a shared project. I also developed practical experience using GitHub in a professional workflow.
Kishani	I gained practical experience in Python programming and problem-solving. I also improved my communication and collaboration skills.I learned how to use GitHub for version control, and gained a better understanding of organizing a project.
Anosh	
Shivantha	I gained some practical experience regarding how to do a group project in a more professional and organised way thanks to GitHub. I learnt some new types of coding functions. Also learnt the importance of being proactive and trying my best not to underestimate/ give less priority to team building activities such as this. The calculator that we ultimately created was a decent one - but it could have been made in a bit more refined and complex way.
Nethushad	I gained practical experience in Python programming and learned how to use GitHub. I also had the opportunity to improve my English communication skills. Furthermore, I learned how to work effectively as part of a team and successfully complete projects together.

Member name	What I gained
Dihini	I gained hands-on experience on python programming and gained practical experience on using GitHub. I also got the chance to improve my English communication skills and learned how to work as a team and do projects.

6. Next Steps

With Project 1 a clear success, the team will now move on to its next project, which will be led and designed by Kishani. It will be a program with a graphical user interface (GUI) that aims to solve a real-world problem.

This project will be demanding, and members will learn many new things. Because it is suitable for inclusion on a CV, members are encouraged to build their Python proficiency beforehand, using practice sites such as Codewars, W3Schools and FreeCodeCamp.